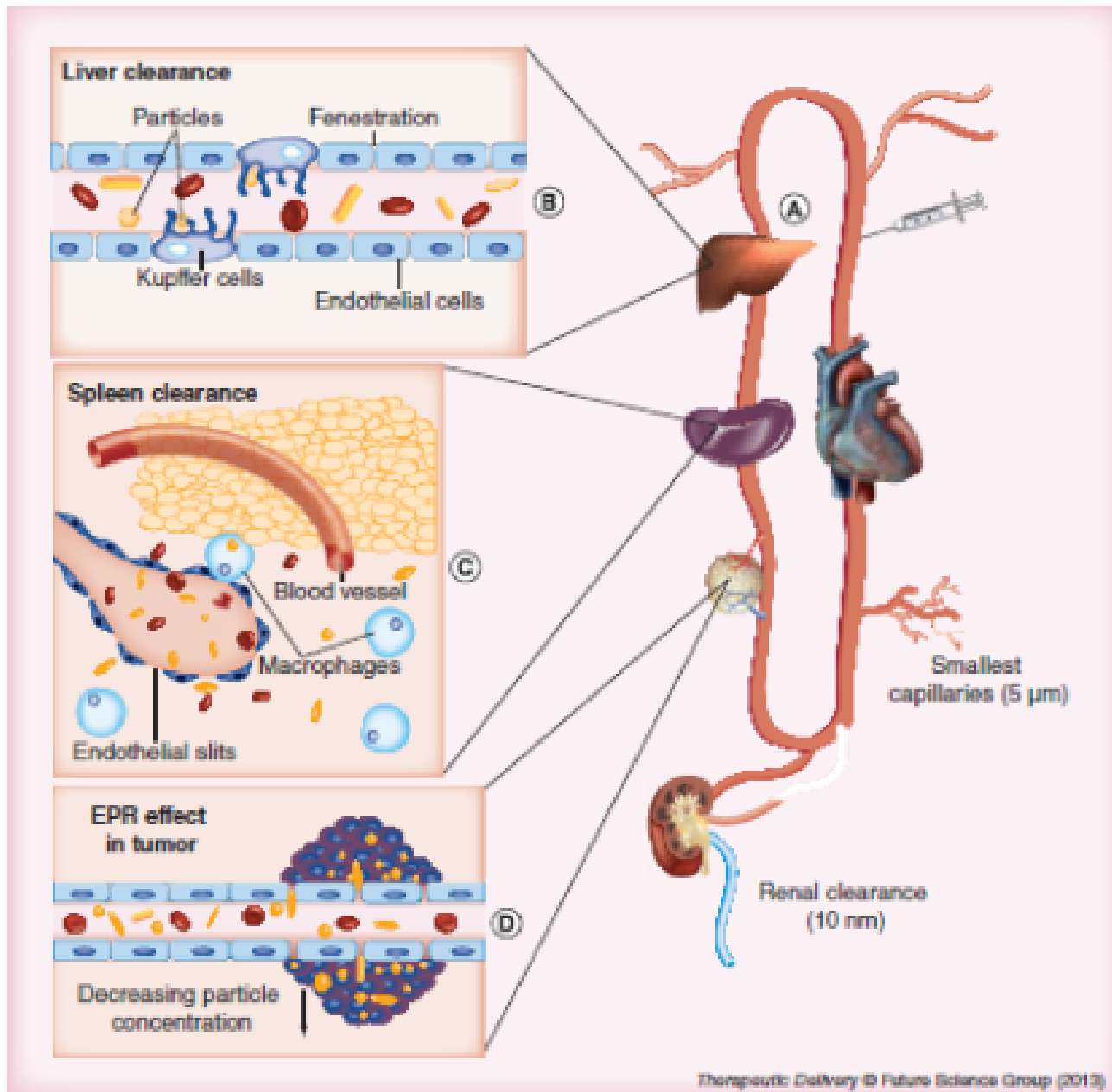
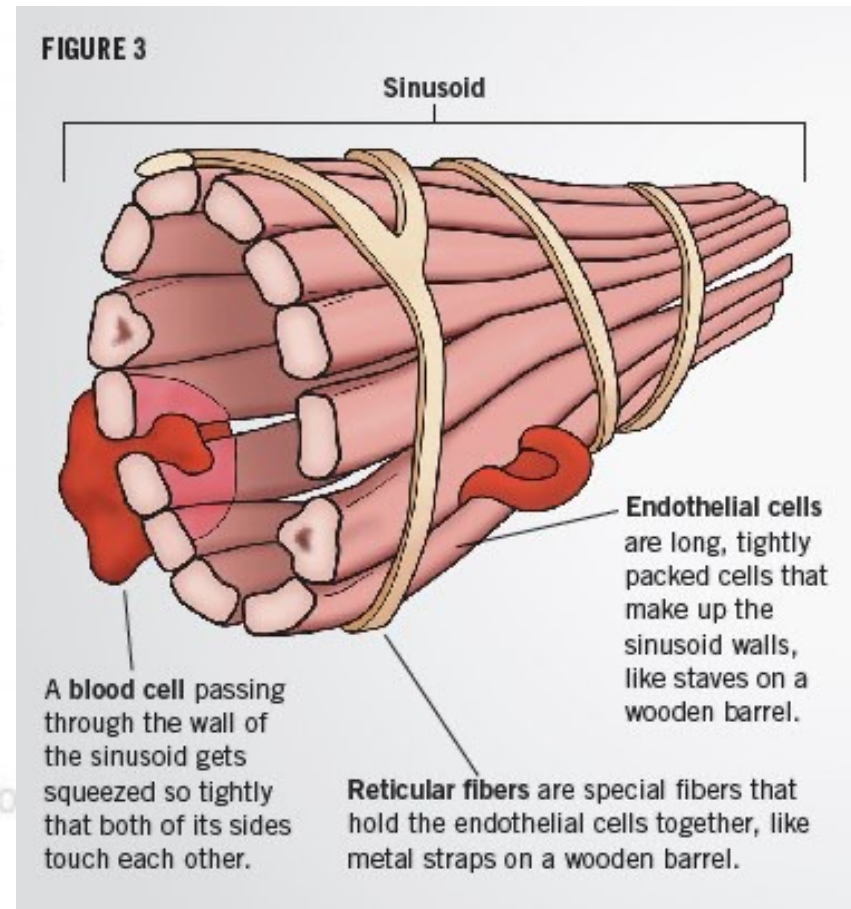
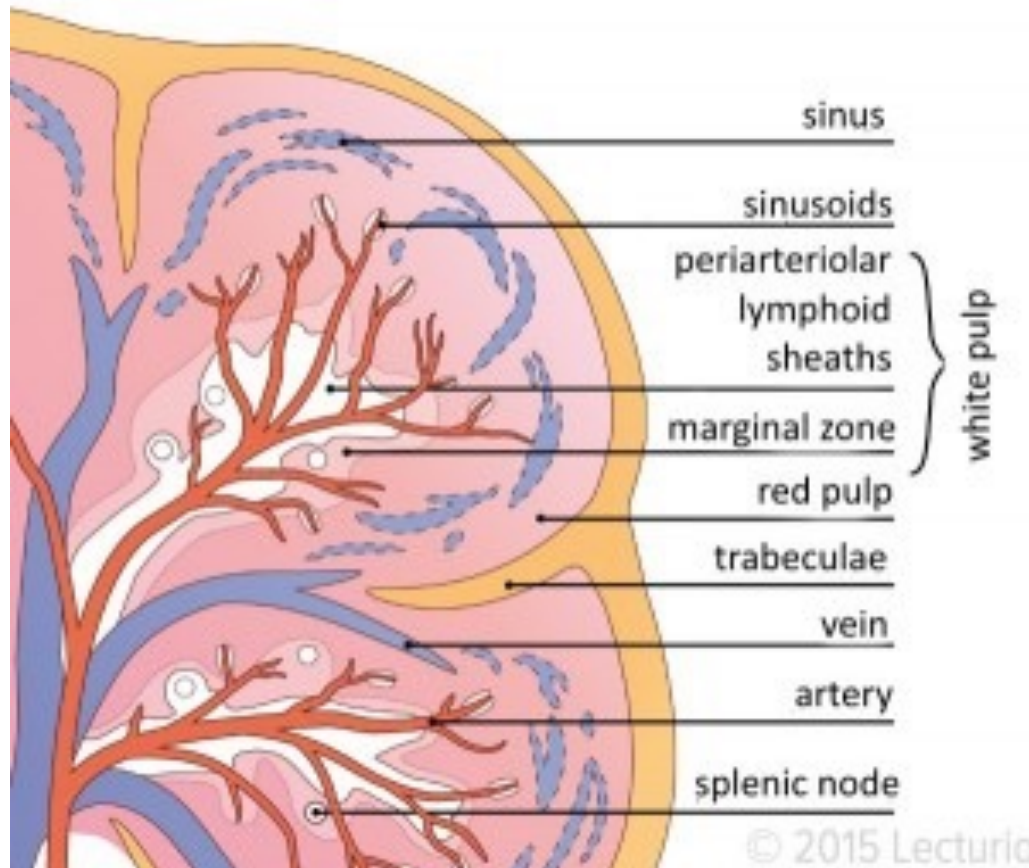


Size requirements



Spleen blood filtration

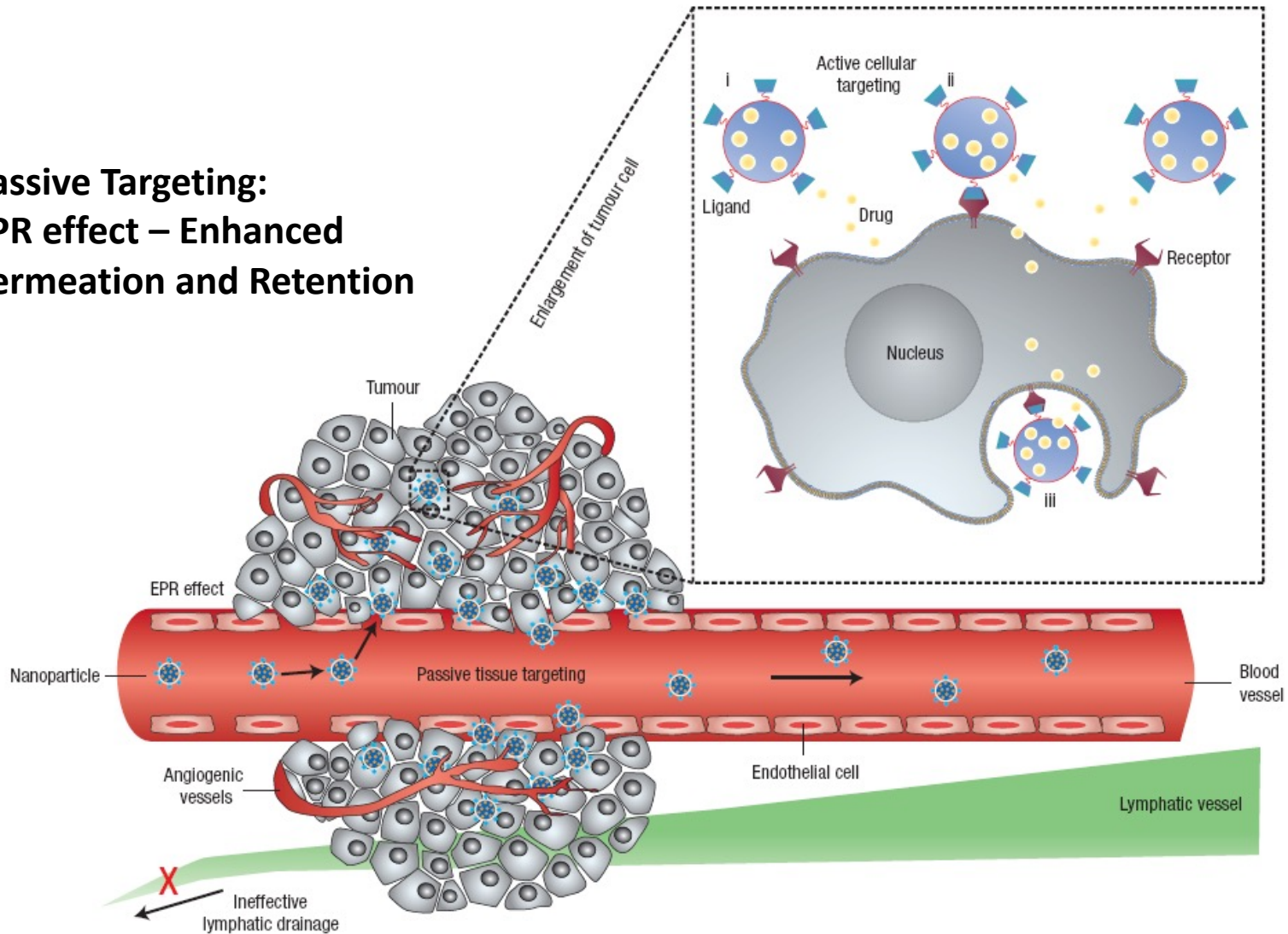


<https://answersingenesis.org/human-body/the-mysterious-spleen/>

<https://www.lecturio.com/magazine/spleen/>

Tumors grow blood vessels that are leaky

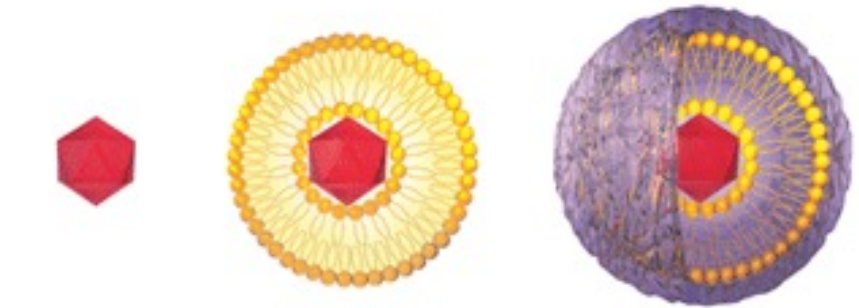
Passive Targeting: EPR effect – Enhanced Permeation and Retention



Approved nanocarriers for cancer: Doxil

Doxorubicin is effective anti-cancer drug but its treatment has high associated cardiotoxicity due to systemic circulation

Limits the dose of Dox that can be administered



Drug: doxorubicin

Chemotherapy agent

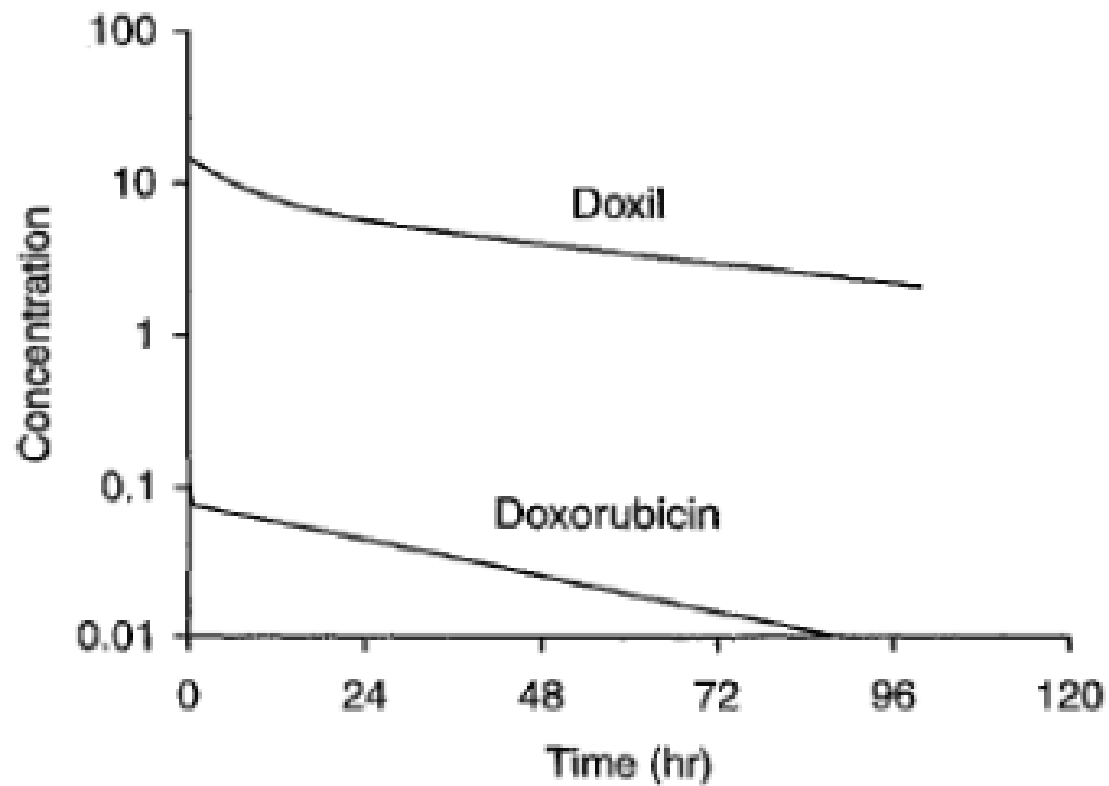
Doxorubicin

Liposome

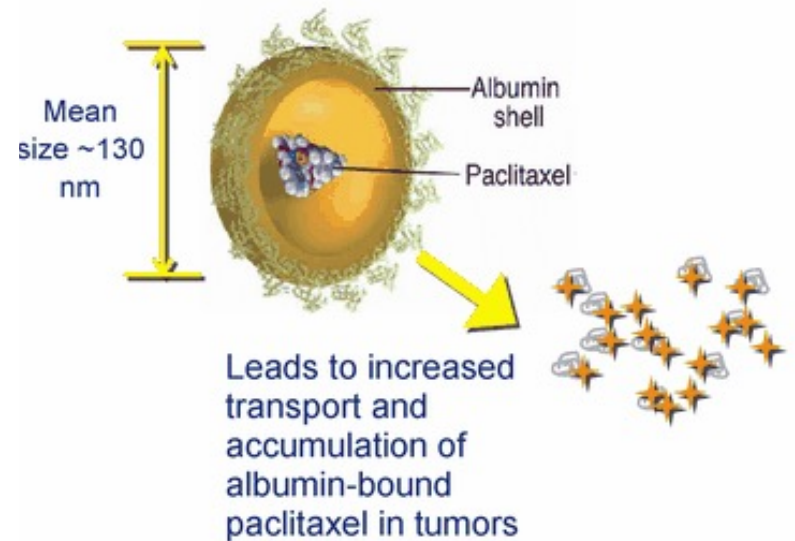
Pegylated Liposome

Name of product: Doxil

Reduced cardiotoxicity



Nano Carriers Example: Abraxane

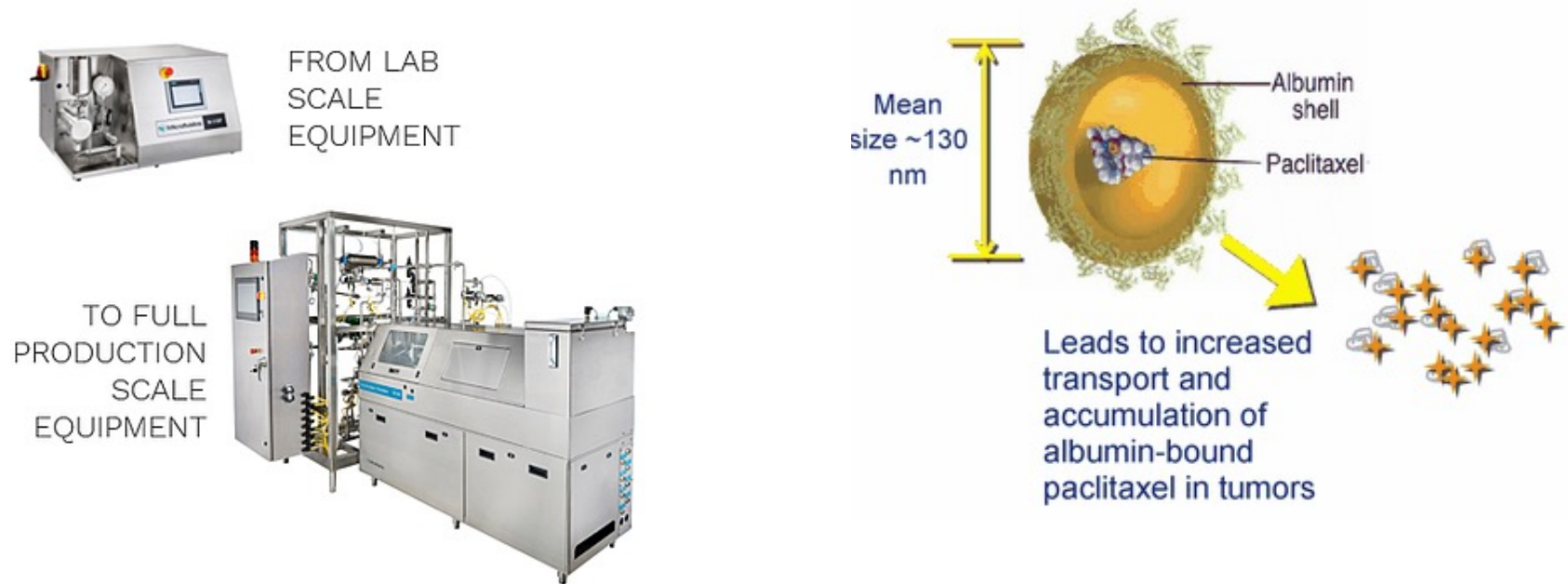


Drug: Paclitaxel

Chemotherapy for breast cancer: Very low solubility in water

Chemotherapeutic bound to protein nano-particle: Albumin bound paclitaxel

Nano Carriers Example: Abraxane



Paclitaxel is dissolved in an organic solvent to form an oil phase. This is then mixed with an aqueous phase – which is the albumin solution. Coarse emulsion which is then processed using high shear to create a nanoemulsion of 130nm.

Name of product: Abraxane

Approved in 2005 (\$134 million in sales that year)

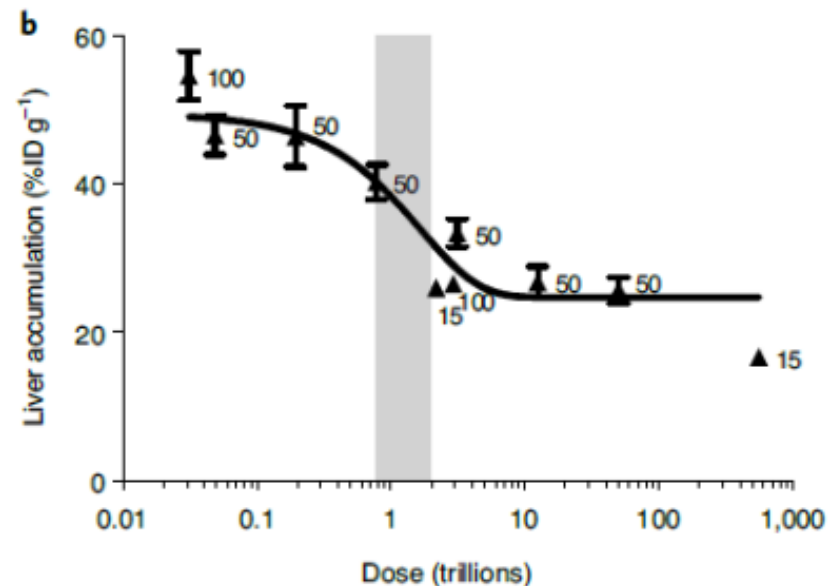
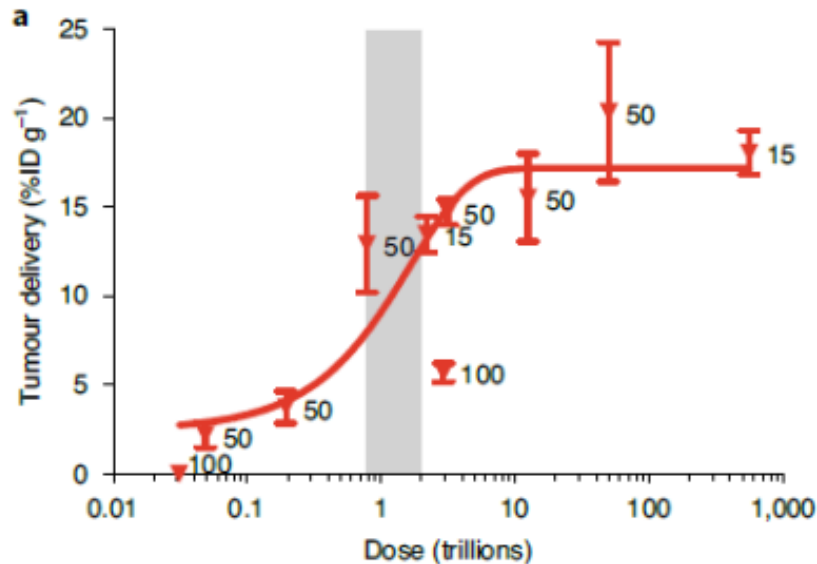
[https://www.microfluidics-mpt.com/applications/nanoparticle-](https://www.microfluidics-mpt.com/applications/nanoparticle-albumin-bound-nab-drug-delivery)

[albumin-bound-nab-drug-delivery](https://www.microfluidics-mpt.com/applications/nanoparticle-albumin-bound-nab-drug-delivery)

<http://www.abraxisbio.com>

EPR effect: challenges

- Only about 0.7% of nanoparticles reach tumor
- Liver is a major sink of IV injected nanoparticles
- Uptake sites in liver can be saturated at high doses that can increase tumor delivery



Numbers indicate particle diameter

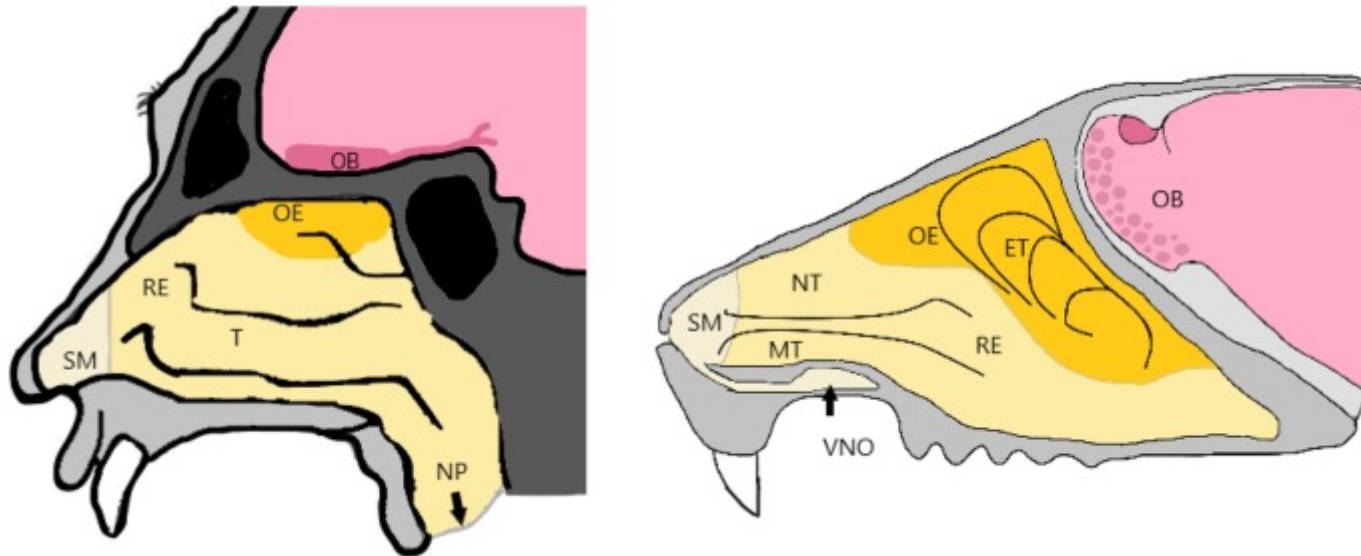
EPR effect: challenges

- Injecting particles through any other route than IV and relying on them to reach systemic circulation is difficult
- Even if they do reach at a low efficiency, you would then expect to only deliver <1% of systemically circulating particles to tumor site.
- Please remember this and not design your group projects like this!

Brain delivery

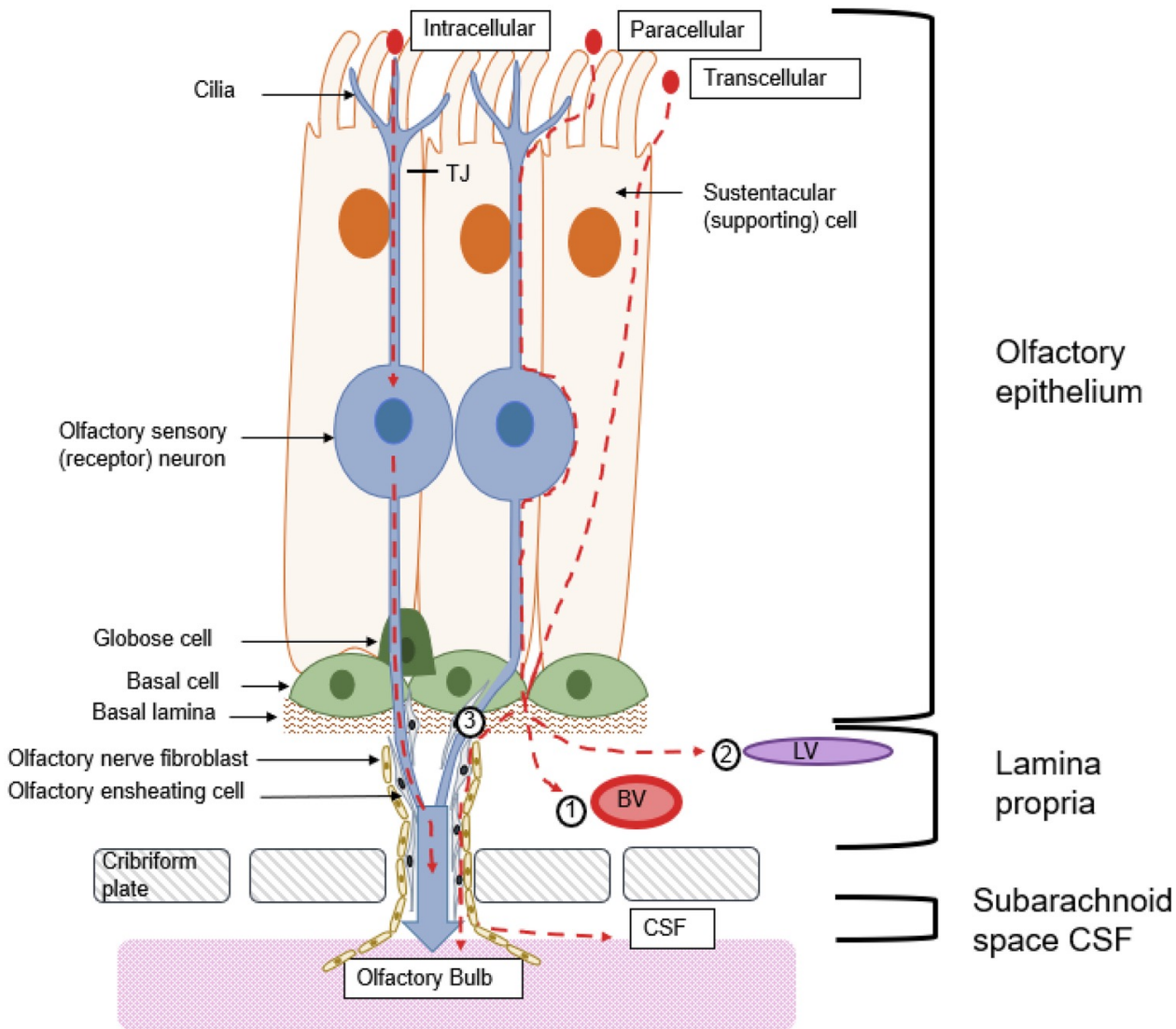
Two possible ways:

1. IV delivery
2. Intra-nasal delivery



Nasal cavity of humans (left) and rodents (right), showing the different regions within the nose. Starting with the squamous mucosa (SM) right at the nostril openings. The respiratory epithelium (RE) covers the main part of the nasal cavity in humans. This is then connected with olfactory epithelium (OE). The olfactory bulb (OB) is in close proximity to the cribriform plate and connected to the OE via the axons of the sensory neurons projecting towards the brain.

Brain delivery through Intra-nasal

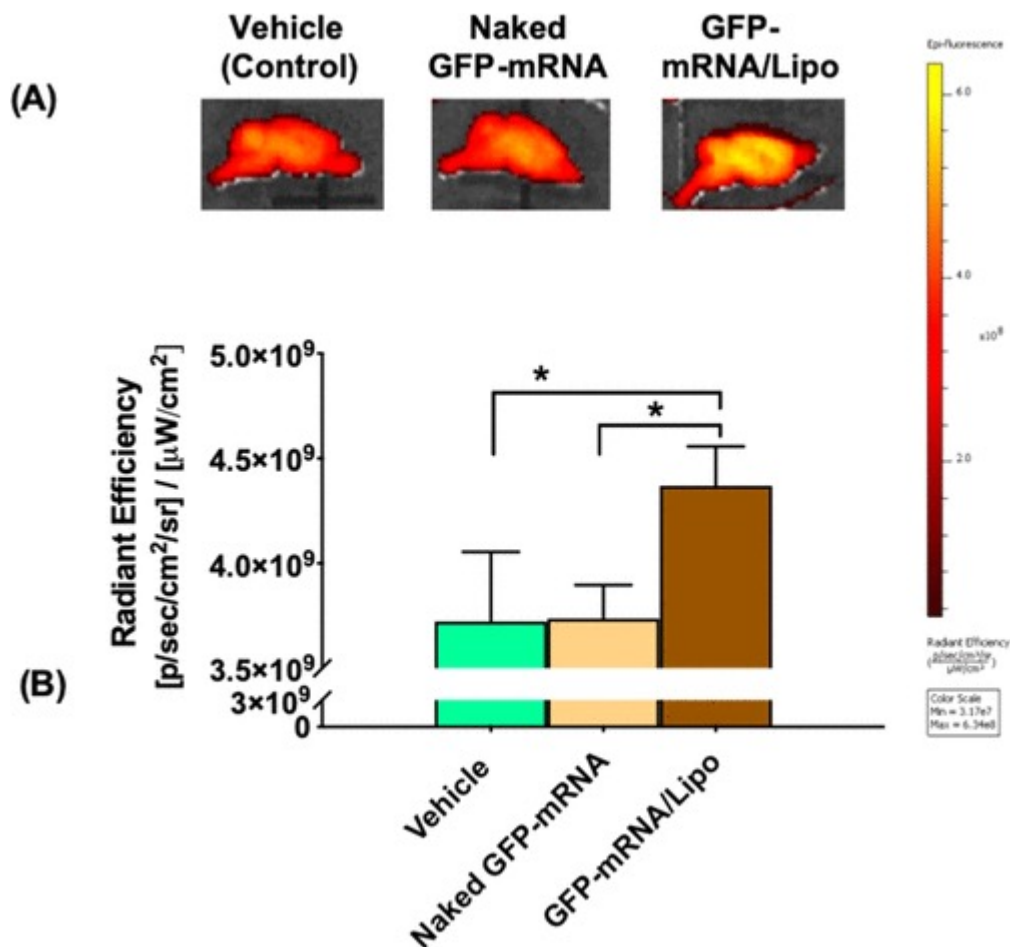


Three different pathways

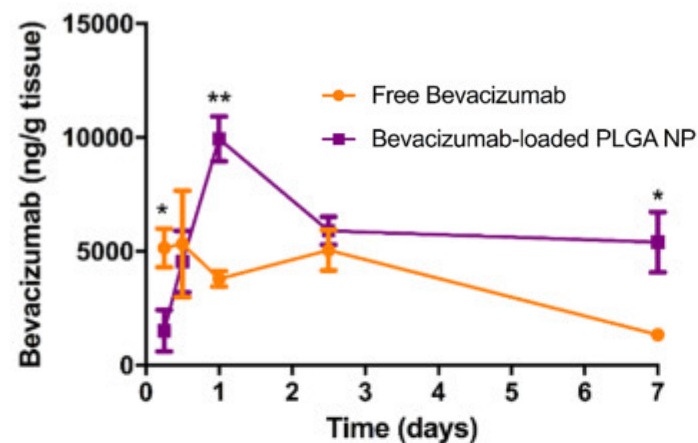
1. The substance can bind to a receptor and be internalised by for example receptor-mediated endocytosis. Then, it travels through the olfactory sensory neuron (OSN) towards the olfactory bulb (OB).
2. Substance uses leaky passages within the OE and travels paracellular into the lamina propria.
3. Substance can also be transported transcellular through the sustentacular cells (SUS) to the lamina propria.

After passing the cribriform plate, the substance can theoretically also reach the cerebrospinal fluid and distribute through the different brain regions

Brain delivery through particles



(A) Brain



200nm PLGA particles

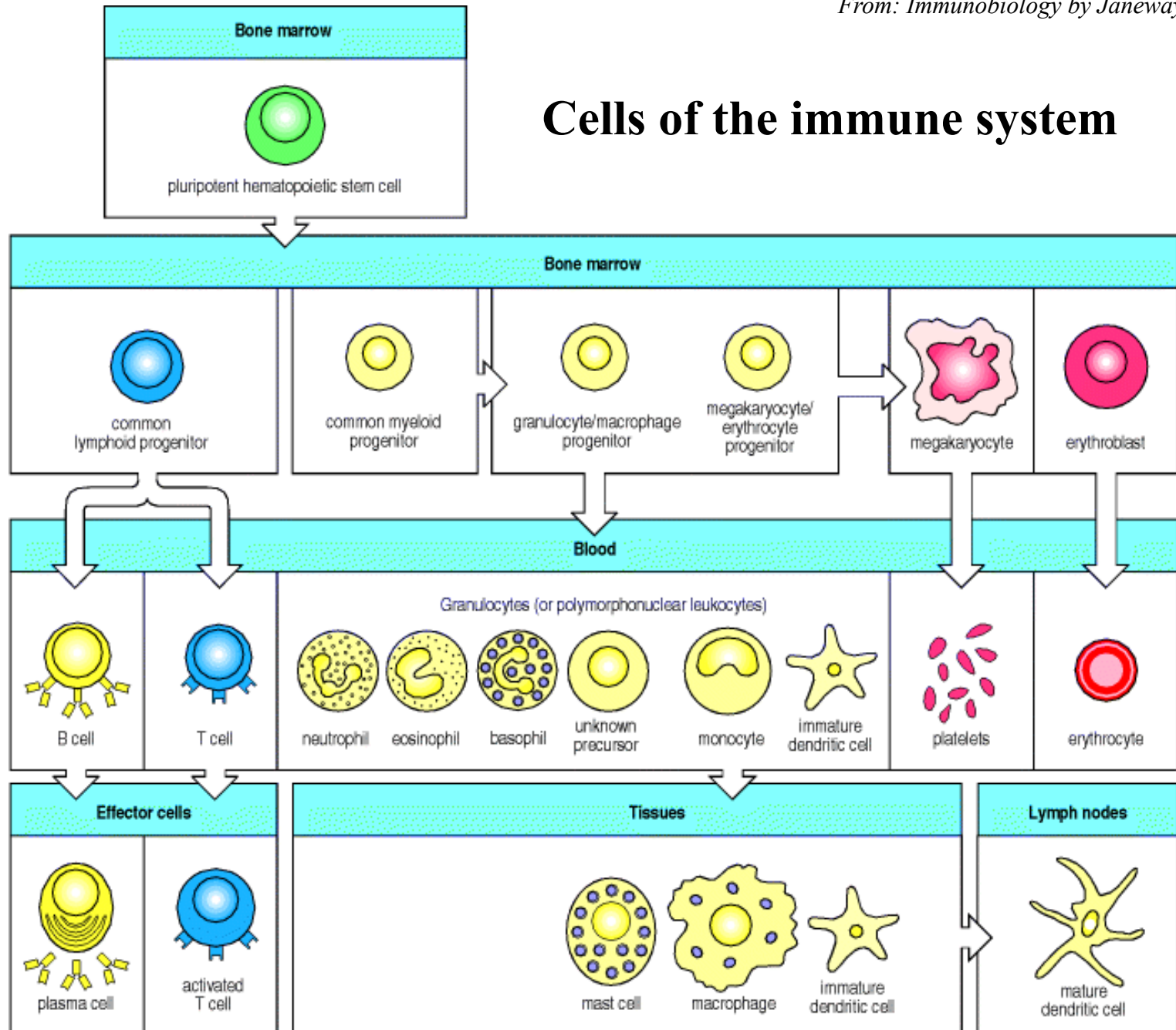
200nm liposomes

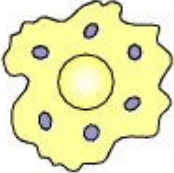
<https://doi.org/10.1016/j.jconrel.2019.07.033>

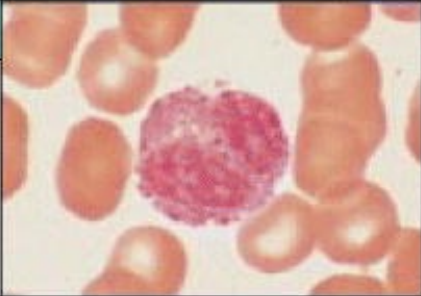
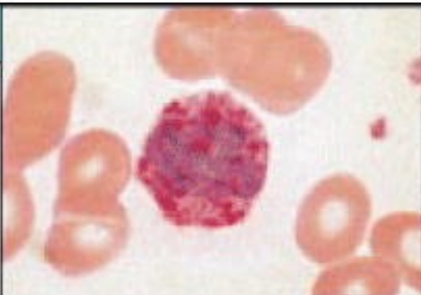
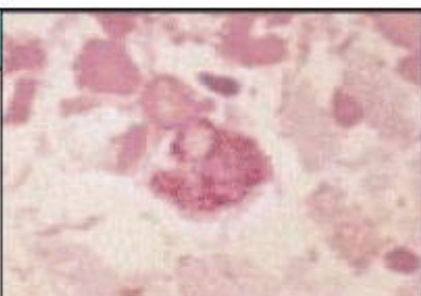
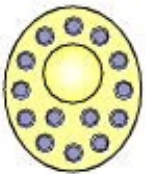
<https://doi.org/10.1021/acs.molpharmaceut.0c00170>

Immune response to materials

Cells of the immune system

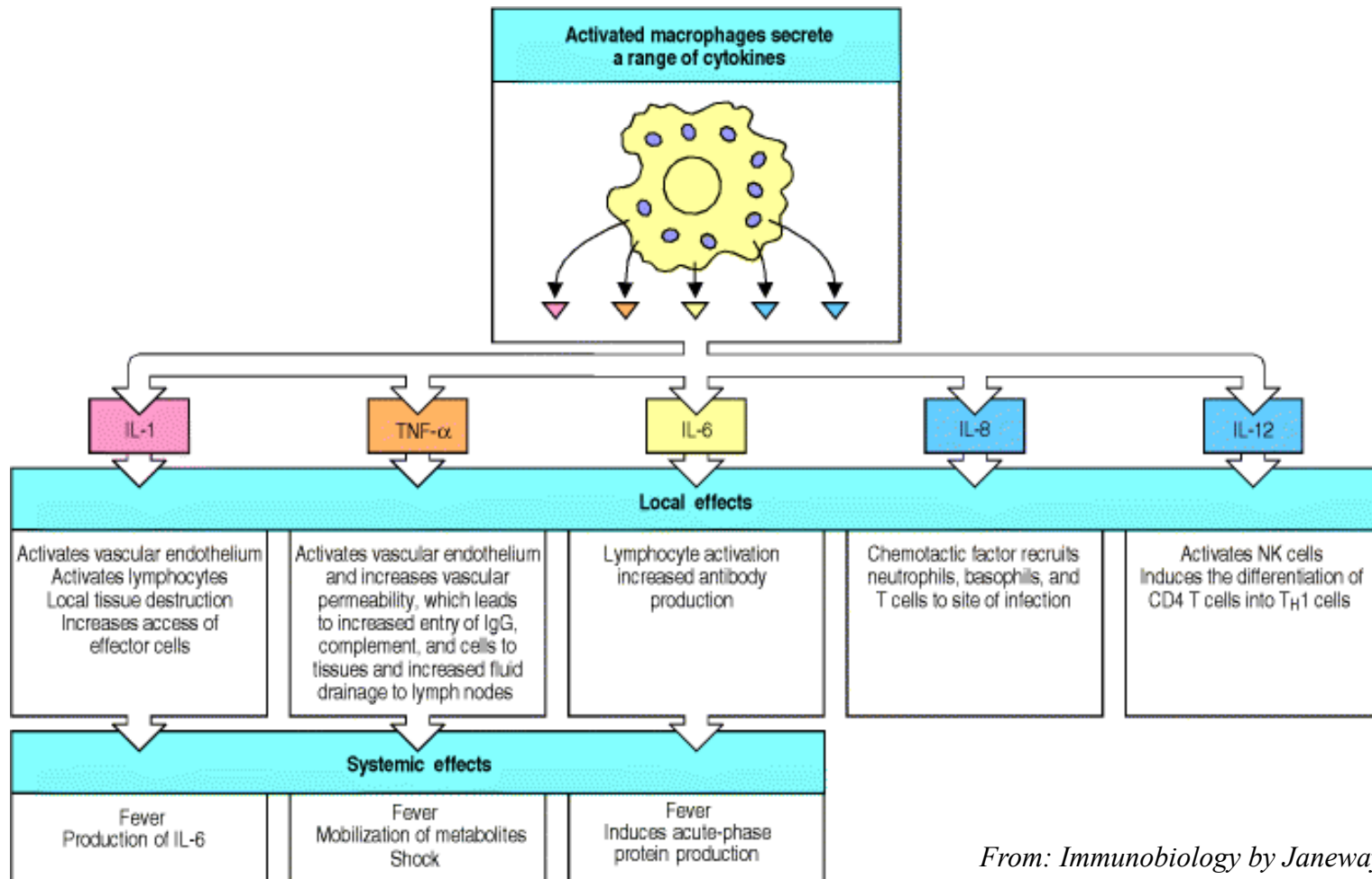


Cell		Activated function
Macrophage		Phagocytosis and activation of bactericidal mechanisms Antigen presentation
		
Dendritic cell		Antigen uptake in peripheral sites Antigen presentation in lymph nodes
		
Neutrophil		Phagocytosis and activation of bactericidal mechanisms
		

Eosinophil		Killing of antibody-coated parasites
		
Basophil		Inflammatory reactions and allergies
		
Mast cell		Release of granules containing histamine and other active agents
		

The Onset of Inflammation

- There are always some phagocytic cells in the tissue at all times
- Upon tissue injury or infection, these cells become active
- Continue digesting foreign Ag and necrotic cells; secrete **CYTOKINES**



From: Immunobiology by Janeway and Travers

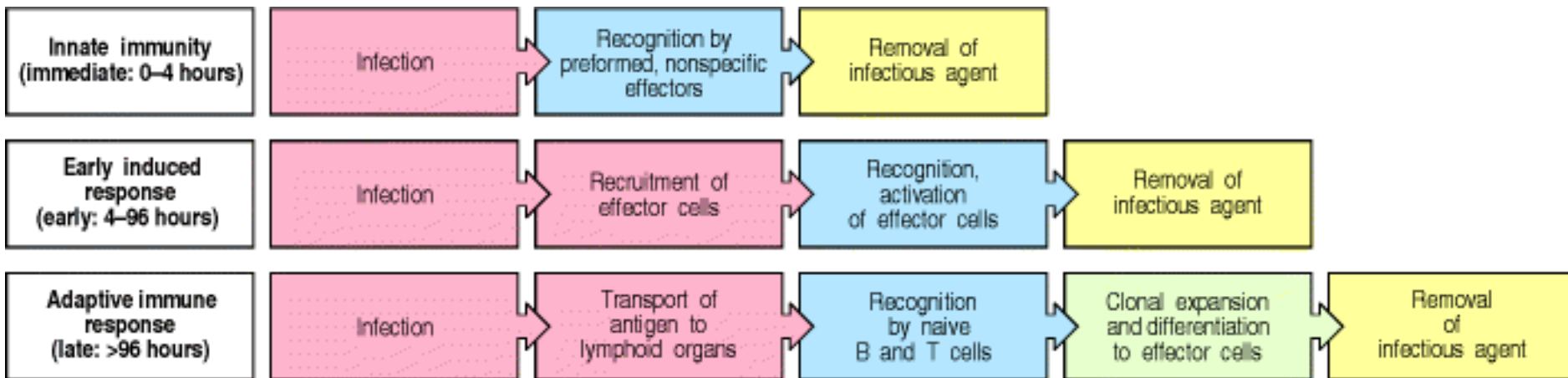
Cytokines and chemokines

- **Cytokines** are small proteins (~25 kDa) that are released by various cells in the body, usually in response to an activating stimulus, and induce responses through binding to specific receptors.
 - They can act in an **autocrine** manner, affecting the behavior of the cell that releases the cytokine,
 - or in a **paracrine** manner, affecting the behavior of adjacent cells.
 - Some cytokines can act in an endocrine manner, affecting the behavior of distant cells, although this depends on their ability to enter the circulation and on their half-life.
- **Chemokines** are a class of cytokines that have chemoattractant properties, inducing cells with the appropriate receptors to migrate toward the source of the chemokine.

Inflammation: Physiological Events

- Tissue injury + foreign entities → activation of tissue macrophages and neutrophils (phagocytic cells) → phagocytosis → cytokine secretion
- Local cytokine release triggers a cascade of events
 - Vasodilation (increase in diameter of the capillaries, redness)
 - Increase in capillary permeability → influx of protein-rich fluid and cells (exudate) → edema (swelling, pain)
 - Influx of leukocytes and phagocytic cells into the tissue
 - Healing, tissue regeneration → formation of new capillaries and scar tissue

Response to an infection



Materials and the immune system

- Main job of the immune system is to protect us from pathogenic invasions
- **Materials in the body → equivalent to foreign invasions**
- The body has employed a range of interactions to counter this invasion
- **Most materials, following implantation or administration, comes in contact with blood**
 - Even if it is not delivered into the blood, the tissue injury caused by implantation causes blood contact
- Degree or intensity of the tissue reaction → Measure of material biocompatibility
- These host reactions are tissue dependent, organ dependent and species dependent

Blood response to materials

- **Complement System**
 - COMPOSED OF **MORE THAN 20 DISTINCT PLASMA PROTEINS** THAT WORK IN AN ORCHESTRATED CASCADE
 - **PURPOSE:** NON-SPECIFIC RECOGNITION AND ELIMINATION OF FOREIGN ELEMENTS FROM THE BODY
- **Blood coagulation on material surfaces**
 - **PHYSIOLOGICAL FUNCTION:** TO STOP BLEEDING FROM INJURED VESSELS, THE TISSUE SURFACE PLAYS AN IMPORTANT ROLE
 - **BIOMATERIAL SURFACES CAN “TRIGGER” THE COAGULATION CASCADE**

Not covering in this course but covered in FoB2

Implications in drug delivery

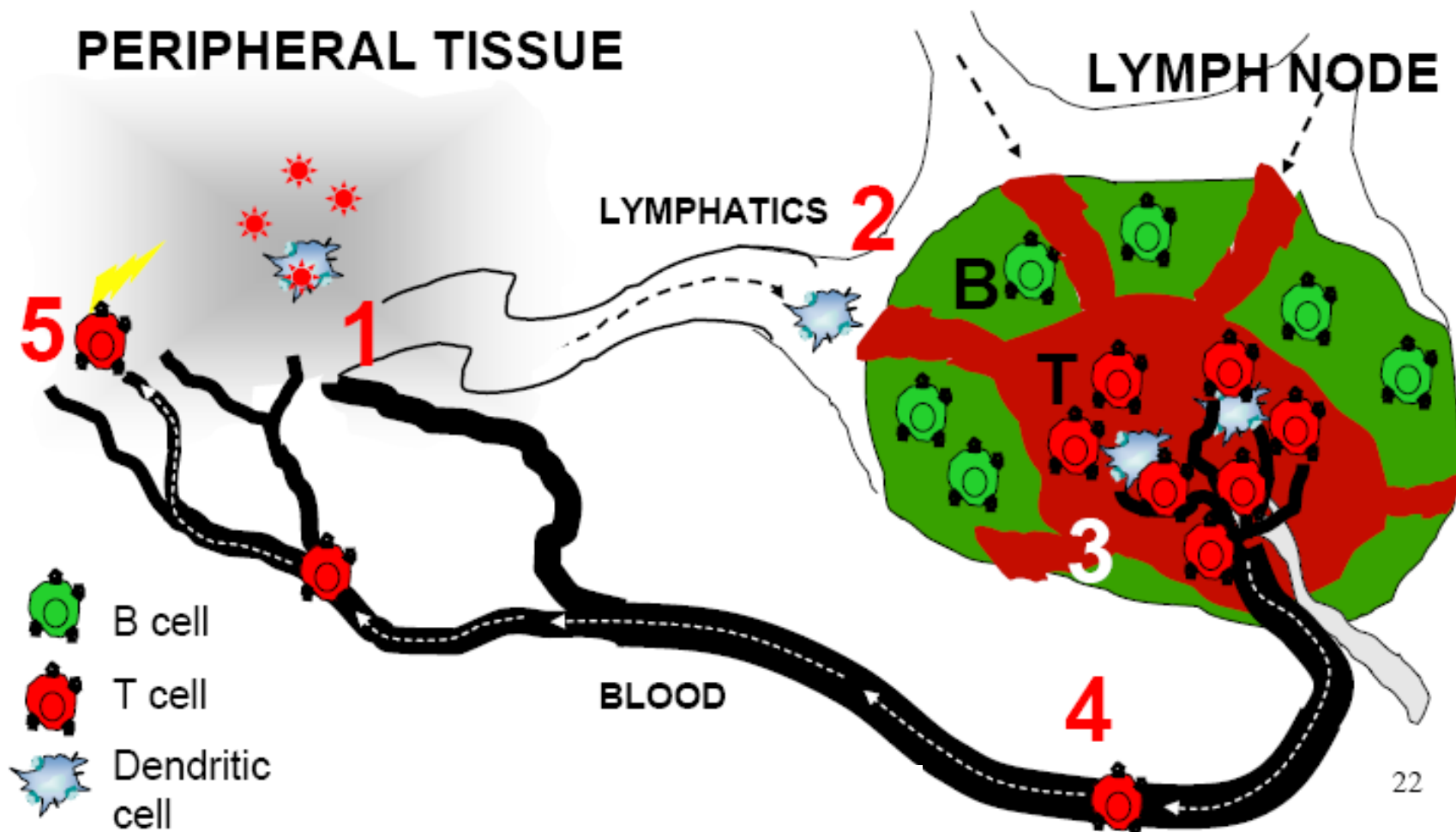
- Effects on drug delivery?
 - Fibrous capsule formation
 - Diffusion limitations: released drug cannot diffuse to the body
 - Limited exchange of fluid
 - Slower rate of hydrolysis and degradation
 - Leads to slower rate drug release
 - Danger to patient

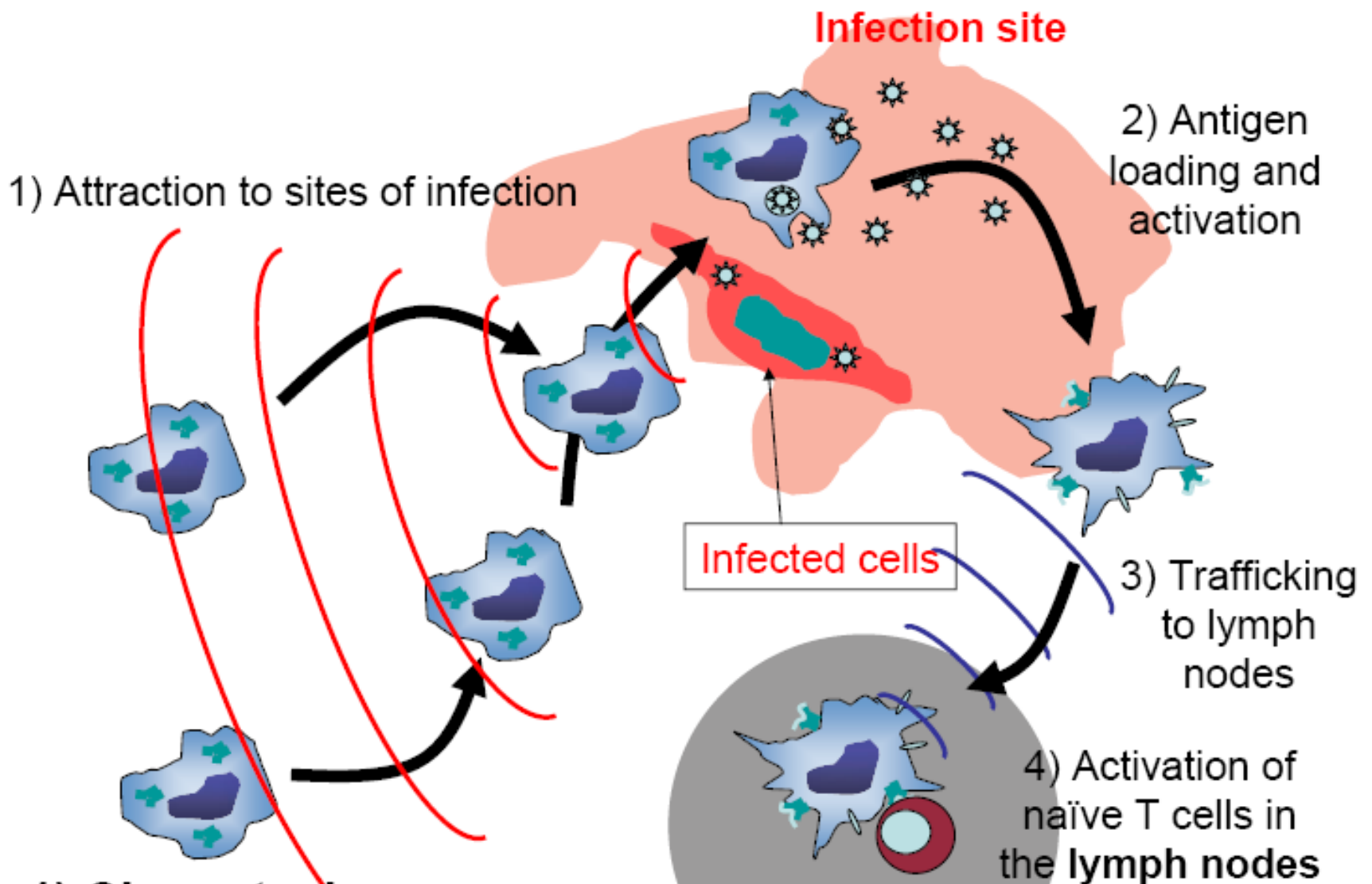
Adaptive Immune Response

Cells of the Adaptive Immune System

- Dendritic cells and macrophages
 - Antigen Presenting Cells
 - Present in the tissue and survey the surroundings
- T cells
 - Helper T cells: Boost and help stronger immune response
 - Cytotoxic T cells: Directly kill the mutant cell
 - Regulatory T cells: Involved in tolerance
- B cells
 - Antigen presentation
 - Antibody generation

Physiology of the primary immune response





1) Chemotaxis:

Migration 'up' concentration gradients of chemoattractant

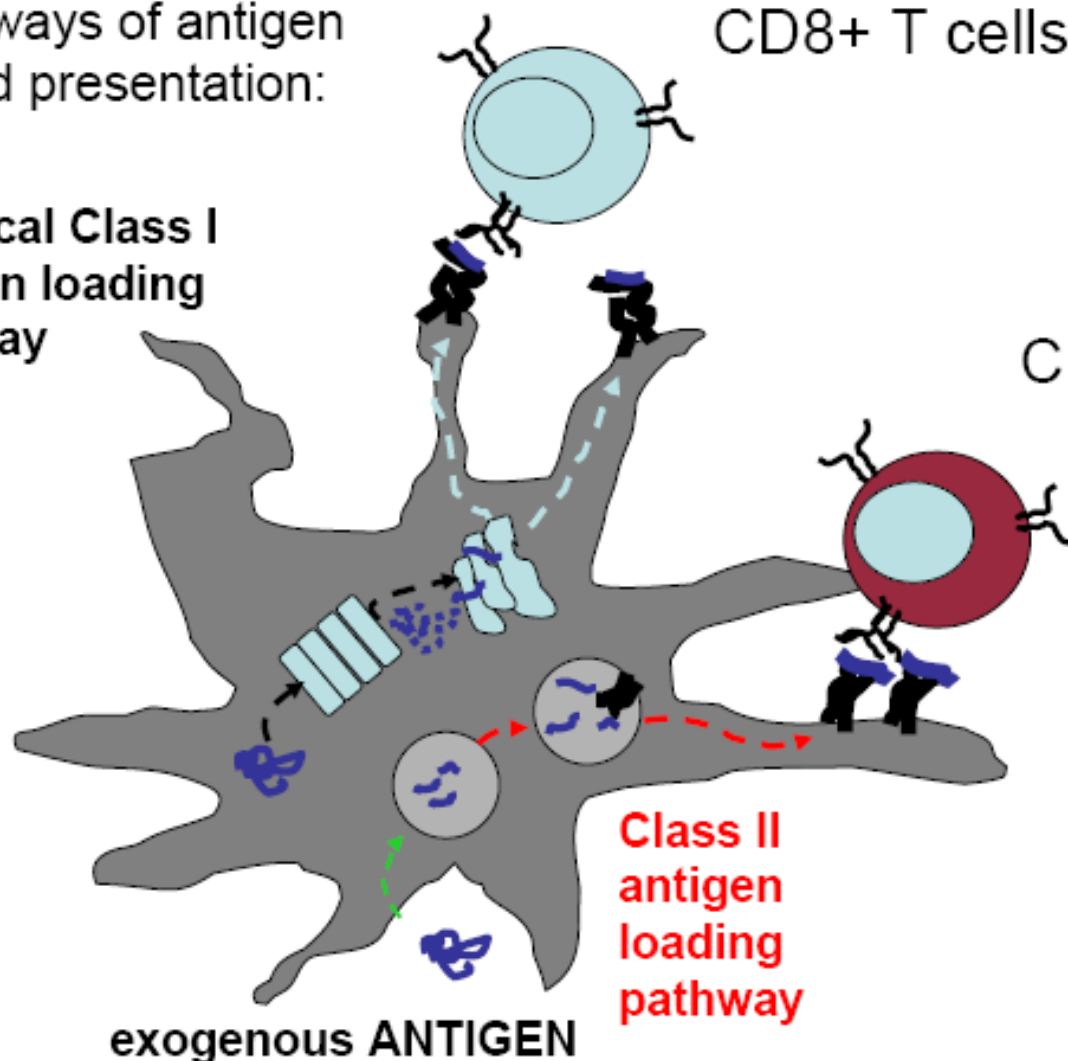
Biology of dendritic cells in T cell activation

Classical pathways of antigen processing and presentation:

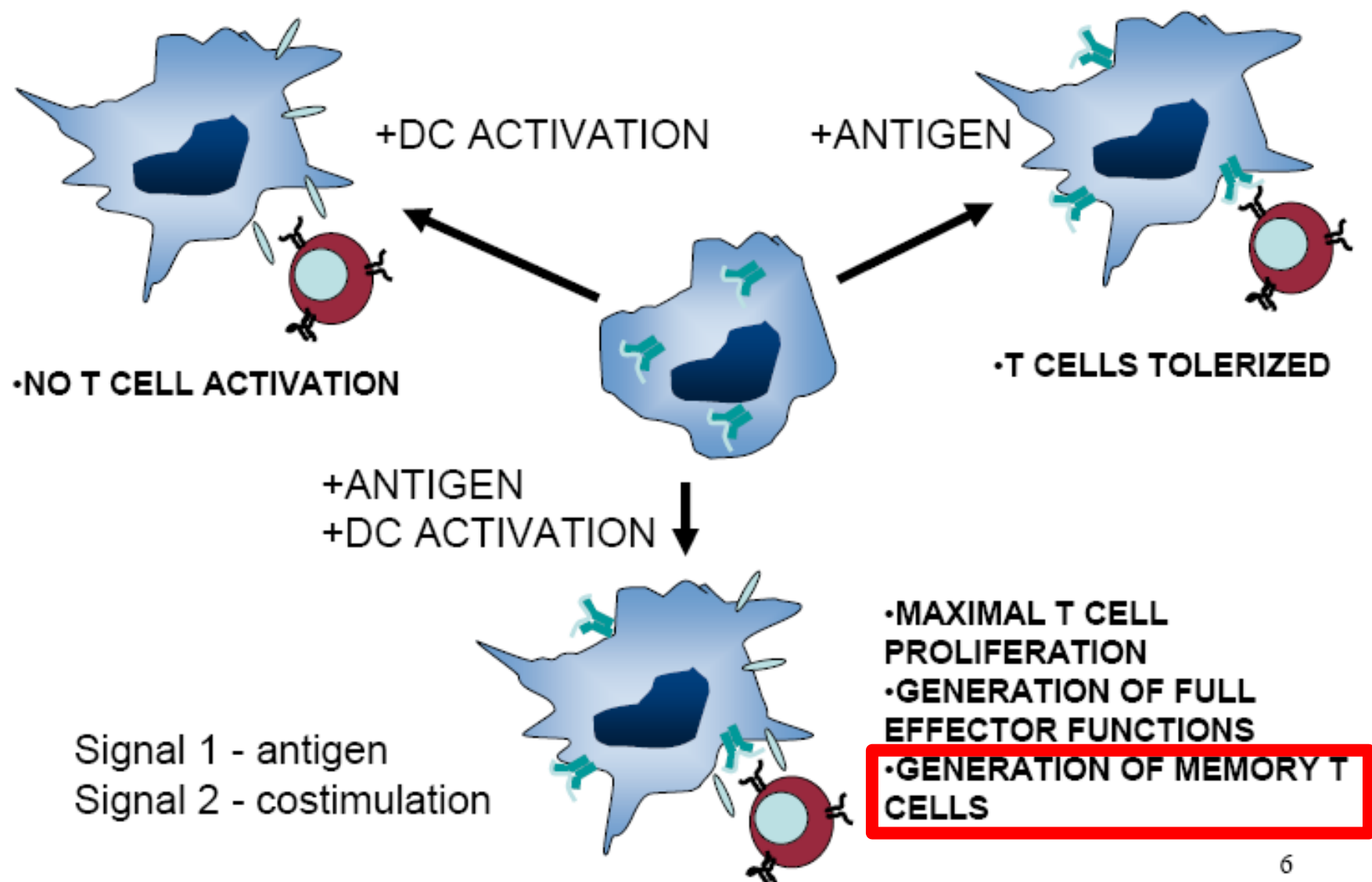
classical Class I
antigen loading
pathway

CD8+ T cells

CD4+ T cells



Antigen is *one* of (at least) *two* signals that must be delivered by a vaccine



Adaptive immune response against biomaterials?

- More relevant for natural polymers (i.e. organism derived proteins, peptides, polysaccharides etc. that the body considers to be “foreign”)
- However, some synthetic polymers can also be recognized as foreign
 - These “polymers” can be processed as “antigens” by dendritic cells and macrophages, presented to T cells which can lead to production of antibodies and cytotoxic T cells.
 - Important criteria while choosing peptides, proteins, polysaccharides etc. as biopolymers for drug and gene delivery.

Immuno-isolated Cell Therapy

- Transplanting cells that can secrete a therapeutic protein over a long period and in response to physiological stimuli
 - Pancreatic Islet Transplantation
 - Long-term gene therapy (using transformed cells)
- The cells transplanted are allogeneic or xenogeneic and can generate immune reaction (rejection)
- A protective polymer encapsulation, that allows nutrients and small molecules (oxygen, electrolytes, nutrients, i.e. < 50 Kd) to reach the cells, but prevents antibody molecules and T cells, → immuno-isolation
 - Replaces constant immuno-suppression
 - Allows xenogeneic transplantation

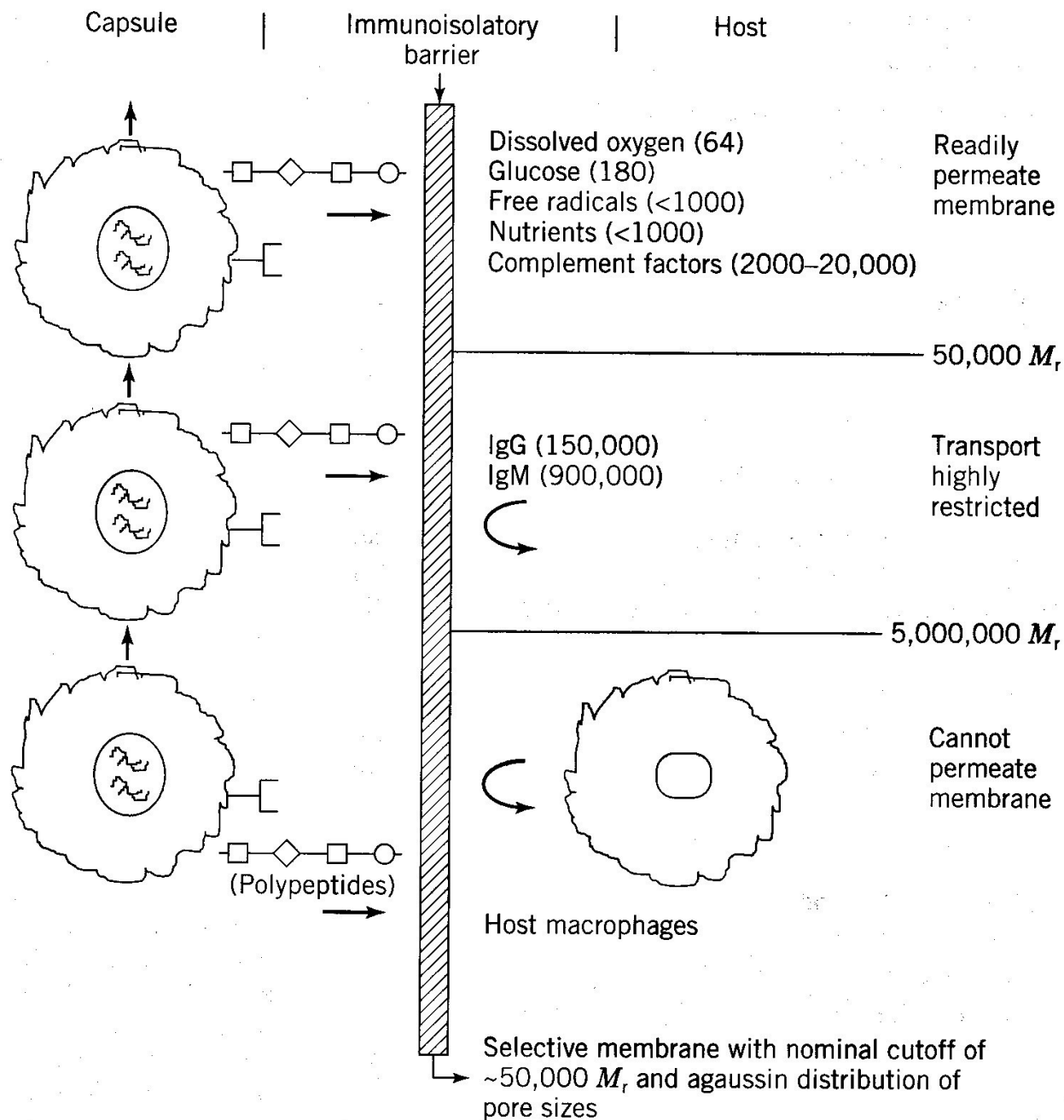


Figure 1. Immunoisolated cell therapy relies upon a membrane to separate foreign secretory cells from a host. Molecules secreted by the cells, usually proteins, can pass through the membrane if they are smaller than $50,000 M_r$. In similar fashion, oxygen glucose and molecules such as insulin can diffuse from the host to the cells. However, the membrane blocks the passage of host cells and restricts transport of host immunoglobulins. Membranes allow passage of complement factors and free radicals that appear to be generated primarily when an implant containing xenogeneic cells is directly implanted into host tissue.

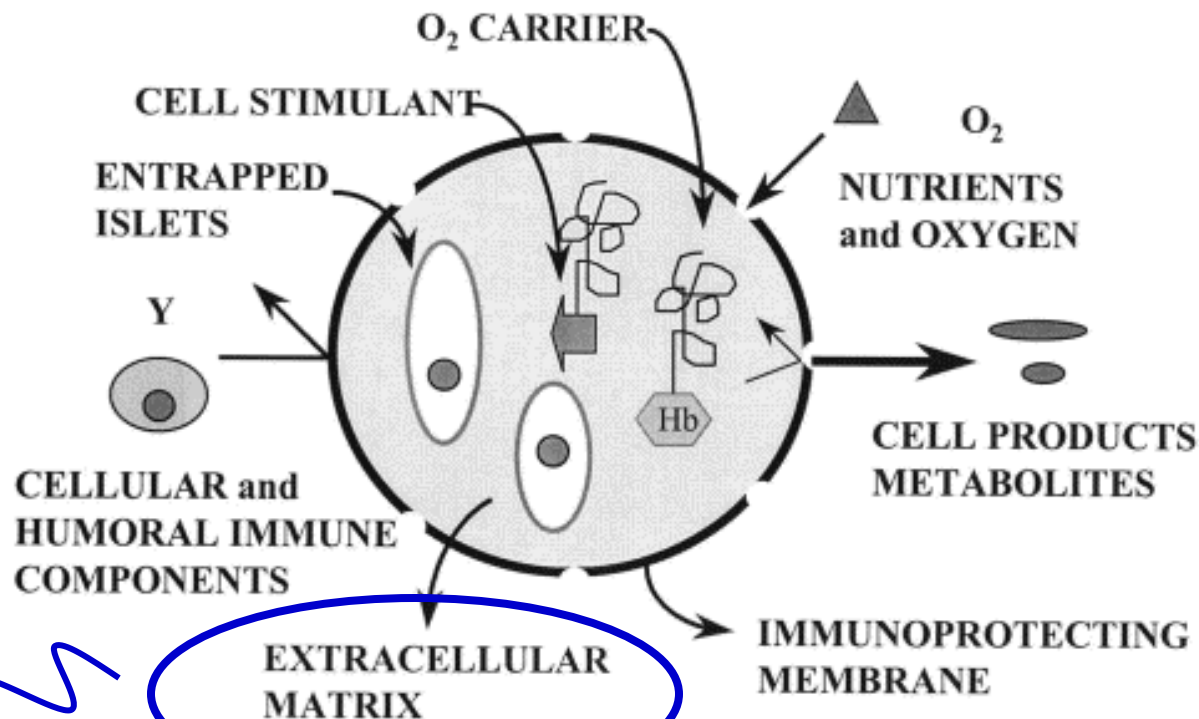
Components of Immuno-isolated Implants

- **Polymeric components:**
 - Must be non-erodible (remains intact over entire transplantation duration)
 - Must be mechanically stable (cannot break)
 - Must be biocompatible (fibrous encapsulation, adhesions etc. Would severely impair diffusion)
 - **Membranes**
 - Hemodialysis membranes: Polysulfone, PAN-VC with cutoff ~ 50,000 daltons → can also be reinforced (cast on a support)
 - **Matrices**
 - Hydrogels: Natural polymers such as alginate or collagen
 - Scaffolds: Polyurethanes coated with hydrophilic polymer

Components of Immunoisulatory Implants

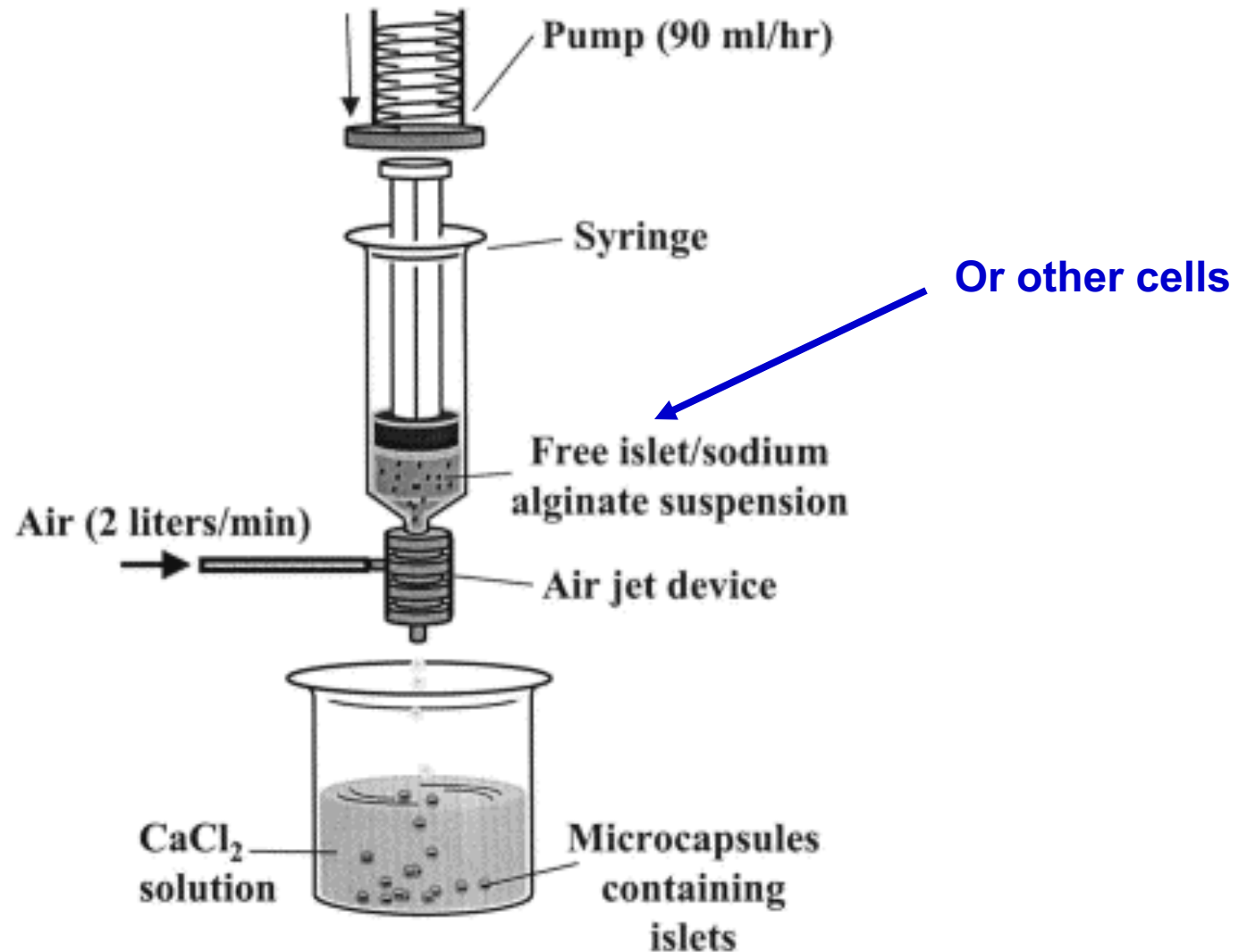
- Cells used for encapsulation
 - Primary
 - Islets (Insulin therapy for diabetes)
 - Hepatocytes (Liver replacement)
 - Cell lines
 - E.g. PC-12 cells secreting dopamine (Parkinson's)
 - Engineered
 - Cell lines transformed with genes encoding for therapeutic proteins like Factor VIII (Hemophilia) or EPO (Anemia)
 - Engineered cells that also contains factors preventing complement activation, apoptosis etc.
 - Chemical “switch” that can be turned on or off by addition of drugs

Model configuration



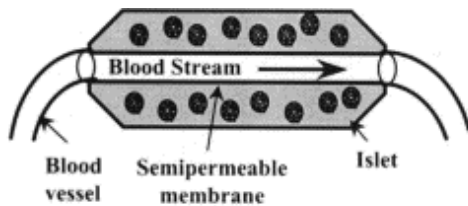
1. Immobilization of the cells to prevent fusion, aggregation and necrosis
2. Improved oxygen transport, such as is provided with perfluorocarbons
3. Presentation of an acceptable physical environment for attachment-dependent cells
4. Provision of an acceptable chemical environment (e.g., collagen gels have been shown to induce epithelial cells to grow into duct-like structures and to promote neonatal endocrine pancreatic cells to organize into islet-like structures)
5. Physical limitation of the cell volume within the chamber
6. Secondary immunoprotection

Schematic of micro-encapsulation process

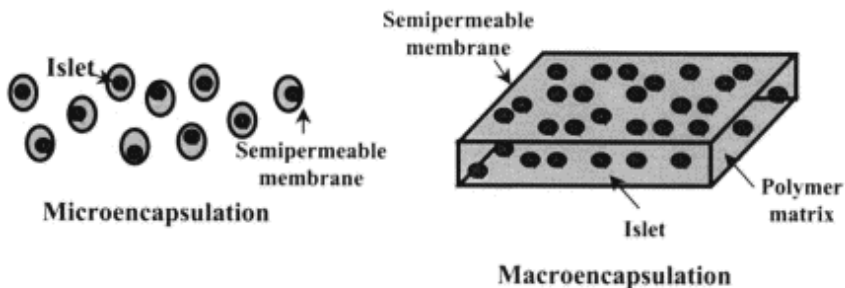


Implant design

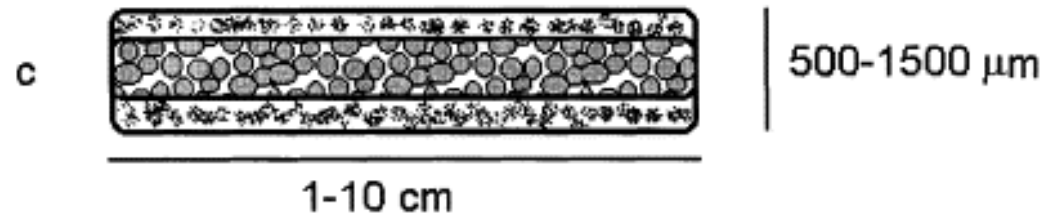
- Grafted cells must be less than 200 to 500 μm away from host environment $\rightarrow$ membrane thickness or matrix properties should be such that oxygen and nutrients can effectively diffuse through



Intravascular Device



Extravascular devices



Microfabricated biocapsules for insulin delivery

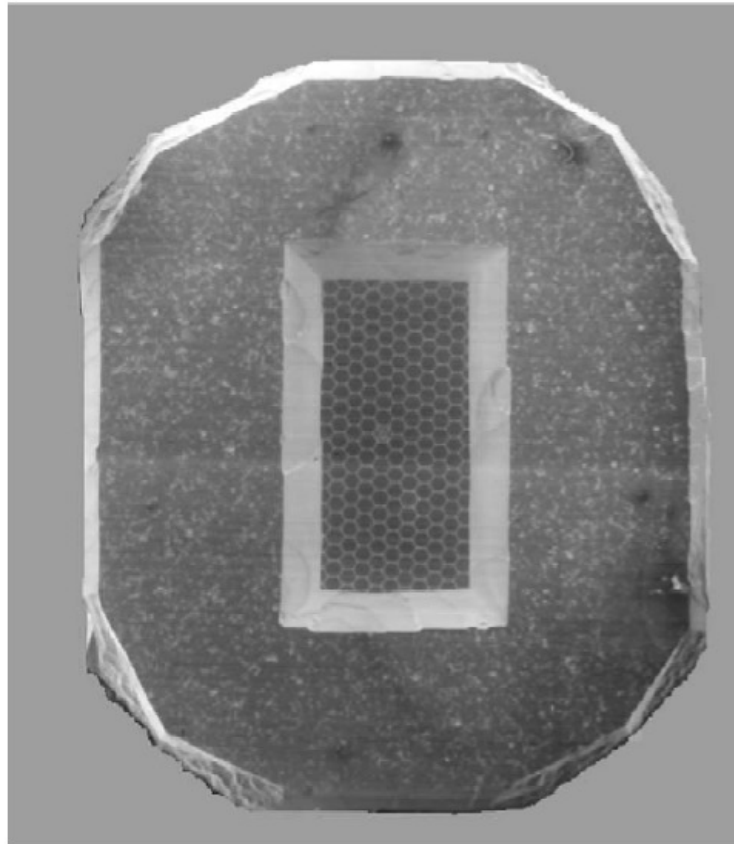


Fig. 7. Micrograph of a biocapsule membrane with 24.5-nm pores.

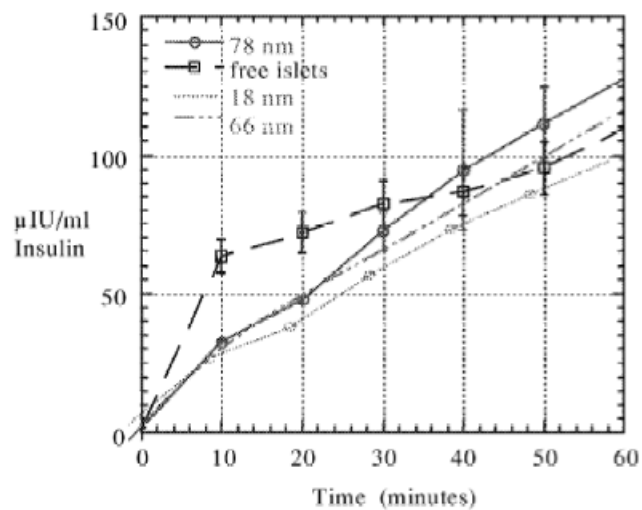


Fig. 8. Insulin secretory profile through differing pore sizes.

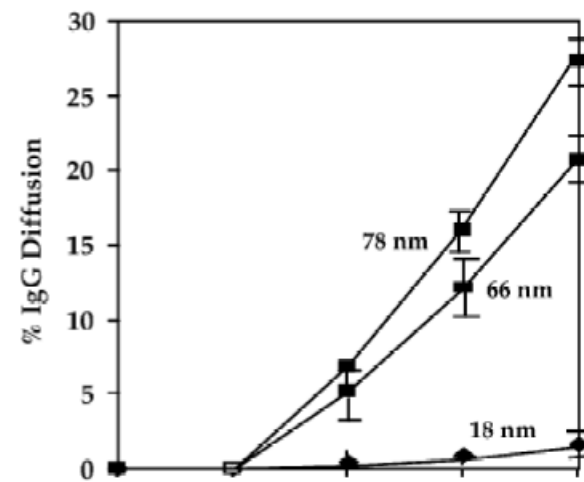


Fig. 9. IgG diffusion through microfabricated biocapsules of three different pore sizes.